AWESOME CYBER SECURITY HEAVEN · REALM DOSSIER · 2026 EDITION

The Proving Grounds

CTF Practice, Labs & Hacking Training

71	131,166	42%	15
FREE TOOLS	COMBINED STARS	ON GITHUB	ACTIVE IN 2026

71 tools · ← Back to the main list · Data verified **2026-10-22** (re-verified monthly)

CURATOR'S NOTES

Legal targets to practice on — vulnerable-by-design apps, CTF platforms, lab ranges. Practice targets are verified live before listing (dead practice apps are worse than none — they teach tool errors, not vulns).

MAINTENANCE SNAPSHOT (2026-10-22)

- Active in the last 30 days: 11 of 29 tracked GitHub repos (38%)
- Quiet classics (no push in 12+ months, still listed): Metasploitable 3, HEVD, APTSimulator, OWASP NodeGoat, flightsim
- Earliest entries: WackoPicko (2010), CryptOMG (2012)

Part of **Awesome Cyber Security Heaven** — 500 curated free tools across 14 realms

FLAGSHIP PICKS — THE HEAVYWEIGHTS

TOOL	★ STARS	CREATED	LAST PUSH	WHAT IT DOES
Vulhub	21.4k	2017	2026-10-17 ✅	Pre-Built Vulnerable Environments Based on Docker-Compose
OWASP Juice Shop	14.2k	2014	2026-10-21 ✅	OWASP Juice Shop: Probably the most modern and sophisticated insecure web application
DVWA	13.9k	2013	2026-10-18 ✅	Damn Vulnerable Web Application (DVWA)
pwntools	13.8k	2013	2026-10-21 ✅	CTF framework and exploit development library
WebGoat	9.4k	2015	2026-10-19 ✅	WebGoat is a deliberately insecure application
GOAD	8.5k	2021	2026-03-12	game of active directory

FULL INVENTORY — 71 CURATED TOOLS

ranked by GitHub adoption · 2026-10-22 census · non-GitHub entries listed last

TOOL	★ STARS	CREATED	LAST PUSH	LICENSE / TIER
Vulhub	21.4k	2017	2026-10-17 ✅	— dockerized vulnerable environments: spin up a CVE, test, tear down. Used by learners spinning up vulnerable environments with Docker; the free compose-file library of real CVE setups. (MIT)
OWASP Juice Shop	14.2k	2014	2026-10-21 ✅	— the modern deliberately-vulnerable app with gamified challenges covering the whole OWASP spectrum. Used in corporate trainings and university courses; the de facto standard vulnerable app for web security practice.

TOOL	★ STARS	CREATED	LAST PUSH	LICENSE / TIER
DVWA	13.9k	2013	2026-10-18 ✅	— the classic Damn Vulnerable Web Application for safely learning injection basics. Used by instructors teaching web vulnerabilities for years; the classic PHP vulnerable app in every classroom lab. (GPL-3.0)
pwntools	13.8k	2013	2026-10-21 ✅	— the CTF exploit-development library: from sockets to ROP chains, pwn everything. Used by CTF players and exploit developers writing exploits fast; the standard Python library for pwn challenges.
WebGoat	9.4k	2015	2026-10-19 ✅	— OWASP's teaching app: guided lessons with a solution walkthrough for each vulnerability class. Used by Java teams and instructors learning web exploitation; the OWASP teaching app with lessons built in. (GPL-2.0)
GOAD	8.5k	2021	2026-03-12	— Game of Active Directory: a three-forest lab purpose-built for safely practicing AD attack techniques. Used by AD red-team trainees practicing real attack paths; the free Game of Active Directory lab commercial courses emulate. (GPL-3.0)
CTFd	6.9k	2015	2026-10-22 ✅	— the CTF platform that powers thousands of events; self-host your own competition. Used by CTF organizers self-hosting competitions; the open-source platform behind thousands of events.
Metasploitable 3	5.8k	2016	2025-02-13 ⏳	— Rapid7's purpose-built vulnerable target VM; the canonical Metasploit practice range. Used by pentest trainees practicing Metasploit workflows; the Rapid7-blessed free practice target.
CloudGoat	3.8k	2018	2026-10-14 ✅	— Rhino Security's "vulnerable by design" AWS scenarios — deploy, attack, tear down. Used by cloud security learners practicing AWS attacks safely; Rhino Security's free vulnerable-by-design deployment.
HEVD	3.1k	2015	2025-02-24 ⏳	— the Windows kernel-driver training target for exploit development. Used by kernel exploit developers learning Windows driver exploitation; the free HackSys training driver with documented vulns. (GPL-3.0)
APTSimulator	2.8k	2018	2025-09-23 ⏳	— makes a Windows host look compromised so you can test detections safely. Used by blue teams generating APT-style artifacts for detection testing; the free Windows simulator for purple exercises. (MIT)
Splunk Attack Range	2.6k	2019	2026-10-20 ✅	— builds instrumented detection labs: attack, collect and test in one workflow. Used by detection engineers building full attack/defend labs in the cloud; the free Splunk range that emulates commercial cyber ranges. (Apache-2.0)
OWASP NodeGoat	2.1k	2013	2024-06-15 ⏳	— the Node.js vulnerable-by-design app: the Top 10 in a controllable lab. Used by Node.js teams learning the OWASP Top 10 in their stack; the OWASP training app for JavaScript. (Apache-2.0)
OWASP Mutillidae II	1.5k	2018	2026-09-24 ✅	— deliberately vulnerable app that re-creates every OWASP Top 10 issue, resettable per install. Used by teachers demonstrating nearly every OWASP vulnerability; the deliberately vulnerable training app. (GPL-3.0)
flightsim	1.4k	2018	2024-04-04 ⏳	— generates malicious network traffic for IDS/NDR validation; defender-side only. Used by defenders safely generating malicious-looking network traffic; the free attack simulator for testing detections. (CC-BY-4.0)
DIVA Android	1.2k	2016	2023-05-19 ⏳	— Damn Insecure and Vulnerable App: the Android training classic. Used by mobile security students learning Android flaw classes; the classic deliberately vulnerable Android app. (GPL-3.0)
DVIA-v2	1.1k	2018	2024-03-29 ⏳	— Damn Vulnerable iOS App v2: the mobile testing curriculum as an app. Used by iOS security learners practicing on Swift; the free deliberately vulnerable iOS app. (MIT)
AzureGoat	967	2022	2024-10-30 ⏳	— the vulnerable-by-design Azure cloud lab. Used by cloud security learners practicing Azure attacks; the free vulnerable-by-design Azure lab. (MIT)

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DSVW	883	2015	2026-08-19	— Damn Small Vulnerable Web: the Top 10 in ~100 lines of Python. Used by trainers demonstrating web flaws with a single Python file; the deliberately small vulnerable app. (public domain)
Vulnerable Node	493	2016	2026-10-06 ✅	— the Node.js vulnerable application for testing analysis tools. Used by Node.js teams auditing a realistic vulnerable app; the free Express-based training target. (MIT-style)
DVWS (Web Sockets)	365	2017	2026-07-24	— the vulnerable WebSocket app from Interference Security. Used by students practicing WebSocket-specific attacks; the vulnerable websockets lab. (MIT)
0l4bs	355	2020	2021-06-02 ⏳	— XSS practice labs with progressive difficulty. Used by students practicing SQLi, XSS and command injection individually; the free isolated labs per bug class. (GPL-3.0)
WackoPicko	354	2010	2024-05-25 ⏳	— the vulnerable app built for testing vulnerability scanners — a frozen-by-design calibration target for your tooling. Used by researchers and instructors demonstrating classic web flaws; the free academic vulnerable social network.
lfi-labs	335	2015	2024-04-08 ⏳	— LFI/RFI/commix practice pages: one flaw family, many variants. Used by learners practicing local file inclusion safely; the free PHP lab for LFI classes. (MIT)
Java Vulnerable Lab	278	2015	2026-10-09 ✅	— the Java vulnerable web application from the CSPF crew. Used by Java developers learning exploitable patterns; the free vulnerable Java application. (GPL-2.0)
DVHMA	274	2015	2018-08-22 ⏳	— the Damn Vulnerable Hybrid Mobile App: Android+iOS appsec in one codebase. Used by mobile security students practicing hybrid-app flaws; the vulnerable hybrid mobile app lab. (Apache-2.0)
CryptOMG	195	2012	2015-06-25 ⏳	— cryptographic flaw testbed from SpiderLabs. Used by crypto learners attacking flawed implementations; the free crypto-focused challenge set. (GPL-3.0)
damnvulnerable.me	34	2015	2019-05-19 ⏳	— a modern vulnerable webapp for the OWASP Top 10 era. Used by mobile security students learning both Android and iOS flaws; the free vulnerable mobile apps. (MIT)
SentinelTestbed	11	2013	2016-11-18 ⏳	— a testbed app for evaluating security instrumentation. Used by engineers testing detection content against a deliberately vulnerable host; the free detection-lab target. (GPL-3.0)
Altoro Mutual	—	—	—	— the classic demo bank: legal target for appsec training and tool demos. Used by security teams demoing scanners on a safe target; the classic IBM testfire-style demo banking site. (free service)
Arizona Cyber Warfare Range	—	—	—	— free live-fire ranges for defenders and researchers. Used by learners attacking live-fire lab environments; the free volunteer-run range. (free service)
bWAPP	—	—	—	— the "buggy web application" with 100+ vulnerability flavors — an old-school playground that still teaches. Used by instructors demonstrating 100+ web bug classes; the free buggy app for classroom labs.
Crackmes.one	—	—	—	— the reverse-engineering challenge community: thousands of crackmes, all legal. Used by reverse-engineering learners practicing on thousands of binaries; the free community crackme archive. (free service)
CryptoHack	—	—	—	— learn modern cryptography by breaking it: challenges from classical ciphers to ECC. Used by learners leveling up applied cryptography; the free interactive challenge platform.
Cryptopals	—	—	—	— the legendary crypto challenges; finish them and AES is no longer magic. Used by engineers learning real-world cryptography; the free challenge set that trained a generation of crypto implementers.

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CTFlearn	—	—	—	— the beginner-friendly CTF platform with a permanent challenge set. Used by absolute beginners solving first flags; the free beginner-friendly CTF platform. (free service)
CTFtime	—	—	—	— the global CTF calendar, team rankings and writeup index — the hub of the sport. Used by CTF teams tracking events and rankings worldwide; the free calendar of every serious competition.
CyberDefenders	—	—	—	— blue-team CTF platform: investigate realistic incidents with the tools a SOC actually uses. Used by aspiring DFIR analysts practicing blue-team cases; the free DFIR CTF platform. (free tier)
ExploitMe Mobile	—	—	—	— the SecurityCompass mobile labs. Used by mobile security learners practicing iPhone and Android labs; the free mobile range. (free service)
Firing Range	—	—	—	— Google's XSS testbed: the original scanner benchmarking target. Used by XSS researchers testing payloads systematically; Google's free testbed for cross-site scripting classes. (Apache-2.0)
Gandalf	—	—	—	— Lakera's free prompt-injection game; learn AI-security attacks hands-on, level by level. Used by teams learning prompt injection hands-on; Lakera's free AI-security game.
Google Gruyere	—	—	—	— Google's cheese-themed vulnerable web app: exploit XSS, XSRF and information disclosure in the cloud. Used by beginners hunting their first web bugs; Google's free vulnerable-app tutorial.
Hack The Box	—	—	—	— the platform for ranked machines, pro labs and active-dir ranges; the industry's proving ground. Used by aspiring OSCP candidates and professional pentesters keeping sharp; the leading freemium training range. (free tier)
Hack This Site	—	—	—	— the community wargames veteran: missions, articles and forums since the early 2000s. Used by a generation of learners starting out; the veteran free training community.
Hackertest.net	—	—	—	— the 20-level hacking simulation for web basics. Used by beginners playing a level-by-level web puzzle; the veteran free challenge series. (free service)
HackThis	—	—	—	— 50+ levels of web security challenges, from basics to advanced. Used by beginners practicing across 50+ mission levels; the free UK challenge community. (free service)
Hackxor	—	—	—	— a realistic web-app hacking game with a story arc — WebGoat with sharper teeth. Used by aspiring web hackers playing a realistic first-person hacking game; the free browser-based range.
LetsDefend	—	—	—	— browser-based SOC simulator: alert queues, playbooks and IR challenges mapped to MITRE ATT&CK. Used by aspiring SOC analysts simulating alert triage; the freemium blue-team range with realistic cases. (free tier)
MASTG Crackmes	—	—	—	— the mobile RE challenges companion to the OWASP MASTG. Used by mobile security learners practicing the OWASP MASTG on real targets; the free crackme suite. (CC-BY-SA-4.0)
Microcorruption	—	—	—	— embedded-security CTF: reverse the lock firmware, level by level. Used by embedded security learners solving lock-firmware puzzles; the free embedded CTF from Matasano lineage. (free service)
MysteryTwister C3	—	—	—	— the international cryptography challenge: from Caesar to quantum. Used by crypto enthusiasts solving graded cipher challenges; the free crypto CTF with legendary pedigree. (free service)
OverTheWire	—	—	—	— Bandit and friends: learn Linux, networking and protocols through SSH wargames. Used by learners mastering Linux and security fundamentals; the classic free wargames every starter list recommends.

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OWASP iGoat	—	—	—	— the iOS learning app from the OWASP mobile family. Used by iOS security learners practicing on Objective-C; the OWASP deliberately vulnerable iOS app. (wiki project)
Pentest-Ground	—	—	—	— hosted practice playground of intentionally vulnerable apps for security testing. Used by testers practicing against free lab environments; the training range from the Netsparker lineage. (free service)
picoCTF	—	—	—	— CMU's beginner-friendly CTF — the best first flag anyone can capture. Used by high-school and university students in CMU's free beginner CTF; the on-ramp for countless security careers.
pwnable.kr	—	—	—	— the classic system-hacking wargame: exploit your way from userland to kernel. Used by exploit developers practicing binary pwn; the classic free challenge server.
Reversing.kr	—	—	—	— the classic Korean reverse-engineering challenge site. Used by RE learners practicing Korean-style binary challenges; the classic free challenge site. (free service)
RingZer0 CTF	—	—	—	— hundreds of challenges across web, crypto, forensics and reversing. Used by learners solving web, crypto and coding challenges; the free online CTF platform. (free service)
Root Me	—	—	—	— hundreds of real-world hacking challenges across dozens of categories, from web to stego. Used by French-speaking and international learners; the free platform with hundreds of hands-on challenges. (free tier)
SmashTheStack	—	—	—	— the wargames network: shellcoding, exploitation and logic puzzles. Used by learners playing classic wargames like Amateria and Blowfish; the veteran free wargame community. (free service)
SocketToMe	—	—	—	— Robin Wood's WebSocket testing challenge app. Used by learners practicing network socket exploitation; the free deliberately vulnerable service. (free service)
techlogia Lab	—	—	—	— free browser-based hacking lab for security training. Used by students practicing end-to-end web exploitation; the free vulnerable lab environment. (CC-BY-4.0)
testasp.vulnweb.com	—	—	—	— the ASP demo target for scanner testing. Used by scanner developers and QA validating detection; Acunetix's free intentionally-vulnerable classic ASP test site. (free service)
testaspnet.vulnweb.com	—	—	—	— the ASP.NET demo target for scanner testing. Used by scanner developers and QA validating detection; Acunetix's free intentionally-vulnerable ASP.NET test site. (free service)
TryHackMe	—	—	—	— guided rooms from your first command to advanced red/blue paths — the friendliest on-ramp. Used by newcomers learning security step by step; the guided-labs platform that made entry-level training free. (free tier)
VulnHub	—	—	—	— downloadable vulnerable VMs; build and break your own home lab offline. Used by students building offline home labs; the free vulnerable-VM library that replaced expensive lab hardware.
Vulnserver	—	—	—	— the TCP vulnerable server for exploit-dev practice. Used by exploit developers practicing buffer overflows and ROP; the standard free TCP exploitation trainer. (free service)
W3Challs	—	—	—	— pentest training challenges with a ranking system. Used by learners practicing web, crypto and stego; the free challenge platform. (free service)
WeChall	—	—	—	— the challenge-site aggregator: one profile across all major wargames. Used by CTF players tracking ranks across challenge sites; the free aggregator for wargame rankings. (free service)

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XSS Game	—	—	—	— Google's XSS tutorial game: six levels from context basics to DOM tricks. Used by beginners learning XSS exploitation; Google's free reverse-engineering-free XSS CTF. (free service)
Zero Bank	—	—	—	— demo banking site for web security exercises. Used by learners practicing web vulns on a simulated banking app; the free demo target behind Invicti training material. (free service)